

Cancellation for the Hessenberg natural sum

Proof of conjecture 00000003522

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Result. The natural sum is cancellative in both arguments. Ordinary ordinal addition fails right cancellation. This is a standard fact about ordinal arithmetic [1], proved here for the supplied assertion.

Definition. For ordinals α, β , their Hessenberg natural sum $\alpha \# \beta$ is recursively the least ordinal strictly greater than every $\alpha' \# \beta$ with $\alpha' < \alpha$ and every $\alpha \# \beta'$ with $\beta' < \beta$. Equivalently, it is the supremum of the successors of those ordinals, taking the supremum of the empty set to be 0. The operation $+$ below always denotes ordinary ordinal addition.

Theorem 1. For arbitrary ordinals α, β, γ ,

$$\begin{aligned}\alpha \# \beta = \alpha \# \gamma &\iff \beta = \gamma, \\ \beta \# \alpha = \gamma \# \alpha &\iff \beta = \gamma.\end{aligned}$$

Nevertheless $0 + \omega = 1 + \omega$ while $0 \neq 1$.

Proof. If $\beta < \gamma$, the defining lower set of $\alpha \# \gamma$ contains $\alpha \# \beta$, so $\alpha \# \beta < \alpha \# \gamma$. Thus the natural sum is strictly increasing in its second argument. Interchanging the roles of the arguments in the same definition shows that it is strictly increasing in its first argument too. For either asserted equality, ordinal trichotomy rules out $\beta < \gamma$ and $\gamma < \beta$, leaving $\beta = \gamma$. The converse follows by substitution.

For ordinary addition, $0 + \omega = \omega$, and continuity at the limit gives $1 + \omega = \sup_{n < \omega} (1 + n) = \omega$. This proves the stated failure of right cancellation. Ordinary addition still satisfies $\alpha + \beta = \alpha + \gamma \Rightarrow \beta = \gamma$; only the opposite direction fails in this example. $\square$

Lean verification and scope. The companion project uses Lean 4.19.0 and the exact Mathlib revision recorded in its manifest. It uses Mathlib’s actual ordinals, natural addition, ordinary addition, and ω . Strict monotonicity follows from the library’s recursive lower-set characterization; injectivity gives both cancellation laws. The final theorem includes the explicit ordinary-addition witness and quantifies over arbitrary ordinals in every Lean universe, with no countability or finite-representation restriction. The library supplies the standard recursive operation; no substitute operation or cancellation axiom is introduced. The formal proof contains no unfinished proof or `native_decide`.

References

- [1] P. Lipparini, *An infinite natural sum*, Mathematical Logic Quarterly **62** (2016), 249–257. See the recursive definition in Section 1 and Proposition 2.2(1). arXiv:1501.05123.